

Customer Churn Analysis for Telecom Industry

Introduction

In the telecommunications industry, retaining customers is just as critical as acquiring them. Customer churn represents a significant loss of recurring revenue. The objective of this project is to analyze customer behavior, identify key drivers behind churn, and build a model to flag high-risk customers. By understanding why customers leave, companies can proactively address issues and deploy targeted retention strategies.

Abstract

This project presents a data-driven approach to predicting customer churn. A dataset featuring demographics and service usage metrics was analyzed. Usage data was aggregated to uncover behavioral patterns. A binary classification machine learning model was developed to predict churn accurately. Model explainability techniques highlighted specific driving reasons, enabling the segmentation of users into actionable categories.

Tools Used

- Python: Used for data processing and modeling tasks.
- Jupyter Notebook: The interactive workspace for executing the code.
- Pandas & NumPy: Crucial for handling and aggregating customer data.
- Scikit-Learn: Used to build and evaluate the classification model.
- ELI5 / SHAP: Libraries used to explain the model's predictions in human-readable terms.
- SQL: Employed for querying transactional data.
- Microsoft PowerPoint: Utilized to compile the final strategic report.

Steps Involved in Building the Project

Step 1: Data Aggregation

SQL and Pandas were used to aggregate critical data points, including total call duration and frequency of support complaints. The data was cleaned to handle missing values and formatted for analysis.

Step 2: Exploratory Data Analysis

Visualizations were created to compare the behavior of customers who churned versus those who stayed. This highlighted significant trends, such as the correlation between high support tickets and churn likelihood.

Step 3: Predictive Model Development

The dataset was split into training and testing sets. A robust machine learning algorithm was trained to recognize subtle patterns that precede a customer's decision to cancel.

Step 4: Model Explainability

Using tools like ELI5, the model's decision-making process was revealed. This step provided actionable insights, showing exactly why a particular customer was flagged as high risk.

Step 5: Customer Segmentation

Based on the model's output, the customer base was segmented into categories like 'At Risk' and 'Loyal'. Strategic recommendations were compiled into a comprehensive presentation.

Conclusion

This project successfully demonstrates how machine learning can transform reactive customer service into proactive customer retention. By accurately predicting churn and explaining the reasons behind it, the project delivers an actionable framework. This empowers telecom companies to focus resources on saving 'At Risk' customers, thereby protecting revenue and improving overall satisfaction.